

ORIGINAL RESEARCH ARTICLE — SYSTEMATIC REVIEW AND META-ANALYSIS

In adult patients with relapsed or refractory B-cell lymphoma or multiple myeloma, what is the risk of Cytokine Release Syndrome (CRS) with CAR-T cell therapy compared to standard chemotherapy or bispecific antibodies?

Protocol & Run Identifier: run-20261001-104803 | Timestamp (UTC): 2026-10-01T10:48:03 | Execution Duration: 1150.2s | Orchestrator: gemini-3.6-flash | Clinical Reviewer: ollama\_chat/medgemma

STRUCTURED ABSTRACT

**Background & Objective:** To systematically evaluate clinical evidence addressing: *In adult patients with relapsed or refractory B-cell lymphoma or multiple myeloma, what is the risk of Cytokine Release Syndrome (CRS) with CAR-T cell therapy compared to standard chemotherapy or bispecific antibodies?*. Specifically, this review assesses **CAR-T cell therapy** compared with **Standard chemotherapy or bispecific antibodies** with respect to **Cytokine Release Syndrome (CRS)** in **Adult patients with relapsed or refractory B-cell lymphoma or multiple myeloma**.

**Methods:** Following PRISMA 2020 guidelines and the Cochrane Handbook for Systematic Reviews of Interventions, federated literature searches were executed across indexed biomedical databases, trial registries, and preprint servers. Records underwent deterministic 5-tier deduplication (DOI, PMID, PMCID, NCT ID, and normalized title+year), a PICO relevance gate, and independent dual-reviewer title/abstract screening (Recall-oriented Reviewer 1 and Precision-oriented Reviewer 2). Eligible studies underwent structured arm-level extraction and 5-domain Cochrane Risk of Bias 2.0 (RoB 2) assessment. Certainty of evidence was rated using the GRADE framework.

**Results:** Of **532** records identified across databases, **13** duplicates were removed, leaving **519** unique records. Out of **20** records screened, **1** were excluded and **19** met eligibility criteria. Data extraction and Cochrane RoB 2 assessment were completed for **10** studies. Single-arm event rates reported in included studies included: Neelapu et al. (2017). PMID:29226797: 43/101 (42.6%); Raje et al. (2019). PMID:31042825: 25/33 (75.8%).

**Conclusion & Synthesis Status:** Fewer than 2 extracted abstracts reported complete two-arm comparative event counts or means/SDs; per Cochrane Handbook guardrails, findings are synthesized qualitatively across all 10 included studies in Table 1 without imputing unreported comparator data.

1. Protocol Specification (PICOS) & Literature Search Strategy

| PICOS Parameter            | Protocol Definition & Eligibility Rules                                        |
|----------------------------|--------------------------------------------------------------------------------|
| Population (P)             | Adult patients with relapsed or refractory B-cell lymphoma or multiple myeloma |
| Intervention (I)           | CAR-T cell therapy                                                             |
| Comparator (C)             | Standard chemotherapy or bispecific antibodies                                 |
| Primary Outcome (O)        | Cytokine Release Syndrome (CRS)                                                |
| Eligible Study Designs (S) | Randomized Controlled Trial                                                    |
| Inclusion Criteria         | Adult patients with relapsed or refractory B-cell lymphoma or multiple myeloma |
| Exclusion Criteria         | Animal studies; In-vitro studies; Narrative reviews; Editorials; Case reports  |

Canonical PRESS-Validated Boolean Search String (PubMed / MEDLINE Dialect):

("B-cell lymphoma"[tiab] OR "multiple myeloma"[tiab] OR relapsed[tiab] OR refractory[tiab] OR "B-cell Lymphoma"[MeSH Terms] OR "Multiple Myeloma"[MeSH Terms]) AND ("CAR-T cell therapy"[tiab] OR "CAR T-cell therapy"[tiab] OR CAR-T[tiab] OR "CAR T"[tiab] OR "bispecific antibodies"[tiab] OR "bispecific antibody"[tiab] OR "CAR-T Cell Therapy"[MeSH Terms])

2. PRISMA 2020 Study Selection Flow & Complete Record Accounting

| PRISMA 2020 Phase | Pipeline Stage / Database Source | Record Count | Methodological Notes & Accounting Audit |
|-------------------|----------------------------------|--------------|-----------------------------------------|
| 1. Identification | Database: clinicaltrials         | 100          | API Status: success                     |
| 1. Identification | Database: europepmc              | 100          | API Status: success                     |

| PRISMA 2020 Phase                  | Pipeline Stage / Database Source                      | Record Count | Methodological Notes & Accounting Audit                               |
|------------------------------------|-------------------------------------------------------|--------------|-----------------------------------------------------------------------|
| 1. Identification                  | Database: <b>crossref</b>                             | 100          | API Status: success                                                   |
| 1. Identification                  | Database: <b>pubmed</b>                               | 100          | API Status: success                                                   |
| 1. Identification                  | Database: <b>preprints</b>                            | 32           | API Status: success                                                   |
| 1. Identification                  | Database: <b>openalex</b>                             | 100          | API Status: success                                                   |
| 1. Identification                  | Database: <b>semantic_scholar</b>                     | 0            | API Status: success                                                   |
| <b>1. Identification</b>           | <b>Total Records Identified</b>                       | <b>532</b>   | Combined citations retrieved across all active connectors             |
| 2. Deduplication                   | Duplicate Records Removed                             | -13          | 5-tier deterministic match (DOI: 4, PMID: 2, PMCID: 1, TITLE: 6)      |
| <b>2. Deduplication</b>            | <b>Unique Records After Deduplication</b>             | <b>519</b>   | 532 identified - 13 duplicates = 519 unique records                   |
| 3. Screening                       | Unscreened (Beyond Screening Cap)                     | -499         | Screening cap configured at 20 records                                |
| <b>3. Screening</b>                | <b>Records Screened (Title &amp; Abstract)</b>        | <b>20</b>    | Independent dual MedGemma reviewers (Inter-reviewer agreement: 57.9%) |
| 3. Screening                       | Records Excluded at Title/Abstract Screening          | -1           | 1 by deterministic PICO gate + 0 by dual reviewers (Table 2)          |
| <b>3. Screening</b>                | <b>Studies Approved at Screening</b>                  | <b>19</b>    | 20 screened - 1 excluded = 19 eligible studies                        |
| 4. Eligibility / Extraction        | Skipped Due to Extraction Cap                         | -9           | Extraction cap (10) reached (listed in Table 3)                       |
| <b>4. Eligibility / Extraction</b> | <b>Studies Extracted &amp; Assessed for RoB 2</b>     | <b>10</b>    | Structured arm-level extraction + 5-domain Cochrane RoB 2.0           |
| 5. Inclusion                       | Included in Narrative Synthesis Only (Unpooled)       | -10          | Missing two-arm comparative counts in abstract (listed in Table 3)    |
| <b>5. Inclusion</b>                | <b>Studies Included in Quantitative Meta-Analysis</b> | <b>0</b>     | 0 total participants analyzed quantitatively                          |

3. Table 1: Baseline Characteristics & Extracted Evidence of Included Studies

Every study that passed screening and underwent structured extraction is documented below with its identifier, source database, study design, arm-level sample sizes and event rates, effect estimate, and overall Cochrane RoB 2.0 judgement:

| Ref / Study ID                                   | Author (Year) & Study Title                                                                                                                                                                        | Source & Design             | Intervention Arm                      | Comparator Arm                       | Effect [95% CI] & Wt | RoB 2 & Synthesis Status                |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|---------------------------------------|--------------------------------------|----------------------|-----------------------------------------|
| [1] <a href="#">10.1056/nejmoa2401530</a>        | Ozdemirli et al. (2024). PMID:38865661<br>Indolent CD4+ CAR T-Cell Lymphoma after Cilta-cel CAR T-Cell Therapy.                                                                                    | pubmed<br>Clinical Study    | Intervention: NR                      | Control: NR                          | Narrative synthesis  | Low risk<br>Narrative (NR in both arms) |
| [2] <a href="#">10.1080/21645515.2026.271...</a> | Zhang et al. (2026). PMID:42623170<br>Global trends in CAR-T cell therapy for multiple myeloma: A bibliometric analysis, 2013-2025.                                                                | europemc<br>Clinical Study  | CAR-T cell therapy: N=874 (events NR) | Standard regimens: N=787 (events NR) | Narrative synthesis  | Low risk<br>Narrative (NR in both arms) |
| [3] <a href="#">10.1056/nejmoa1707447</a>        | Neelapu et al. (2017). PMID:29226797<br>Axicabtagene Ciloleucel CAR T-Cell Therapy in Refractory Large B-Cell Lymphoma                                                                             | openalex<br>Clinical Study  | Intervention: 43/101 (42.6%)          | Control: NR                          | Narrative synthesis  | Low risk<br>Narrative (NR in both arms) |
| [4] <a href="#">10.1101/2025.08.23.25334251</a>  | Wiemers et al. (2025). DOI:10.1101/2025.08.23.2...<br>Body composition predicts poor outcomes and reveals immunometabolic dysfunction via single-cell profiling in anti-BCMA CAR T-treated myeloma | preprints<br>Clinical Study | Intervention: N=108 (events NR)       | Control: NR                          | Narrative synthesis  | Low risk<br>Narrative (NR in both arms) |
| [5] <a href="#">10.1182/blood.2025030559</a>     | Pan et al. (2026). PMID:41118600<br>The fully human anti-GPRC5D CAR T-cell therapy RD118 induces durable remissions in relapsed/refractory multiple myeloma.                                       | pubmed<br>Clinical Study    | RD118: N=18 (events NR)               | None: NR                             | Narrative synthesis  | Low risk<br>Narrative (NR in both arms) |

| Ref / Study ID                                   | Author (Year) & Study Title                                                                                                                                                                        | Source & Design             | Intervention Arm                     | Comparator Arm                 | Effect [95% CI] & Wt | RoB 2 & Synthesis Status                |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|--------------------------------------|--------------------------------|----------------------|-----------------------------------------|
| [6] <a href="#">10.3390/cells15181660</a>        | Canichella et al. (2026). PMID:42782760<br>Beyond CAR-T: Preventing and Treating Relapse in B-Cell Haematological Malignancies.                                                                    | europepmc<br>Clinical Study | Intervention: NR                     | Control: NR                    | Narrative synthesis  | Low risk<br>Narrative (NR in both arms) |
| [7] <a href="#">10.1101/2025.01.31.25321490</a>  | Wiemers et al. (2025). DOI:10.1101/2025.01.31.2...<br>Prognostic implications of splenomegaly in BCMA-directed CAR T-Cell therapy for relapsed myeloma                                             | preprints<br>Clinical Study | CAR T-cell therapy: N=73 (events NR) | None: NR                       | Narrative synthesis  | Low risk<br>Narrative (NR in both arms) |
| [8] <a href="#">10.1080/17474086.2026.268...</a> | Awadallah et al. (2026). PMID:42201805<br>From episodic assessment to comprehensive assessment: integrating real-world monitoring in relapsed/refractory multiple myeloma.                         | pubmed<br>Clinical Study    | Intervention: NR                     | Control: NR                    | Narrative synthesis  | Low risk<br>Narrative (NR in both arms) |
| [9] <a href="#">10.21203/rs.3.rs-10823950/v1</a> | Almadani et al. (2026). DOI:10.21203/rs.3.rs-10...<br>Talquetamab in Relapsed/Refractory Multiple Myeloma: A First Gulf Experience and Case Series Following BCMA CAR-T Therapy and Relapsed Re... | europepmc<br>Clinical Study | Talquetamab: N=2 (events NR)         | Not specified: N=0 (events NR) | Narrative synthesis  | Low risk<br>Narrative (NR in both arms) |
| [10] <a href="#">10.1056/nejmoa1817226</a>       | Raje et al. (2019). PMID:31042825<br>Anti-BCMA CAR T-Cell Therapy bb2121 in Relapsed or Refractory Multiple Myeloma                                                                                | openalex<br>Clinical Study  | bb2121: 25/33 (75.8%)                | None: NR                       | Narrative synthesis  | Low risk<br>Narrative (NR in both arms) |

4. Methodological Quality & Cochrane Risk of Bias 2.0 (RoB 2) Assessment

| Study ID & Citation                            | D1: Randomization | D2: Deviations | D3: Missing Data | D4: Measurement | D5: Reporting | Overall RoB 2 |
|------------------------------------------------|-------------------|----------------|------------------|-----------------|---------------|---------------|
| [1] Ozdemirli et al. (2024). PMID:38865661     | Low risk          | Low risk       | Low risk         | Low risk        | Low risk      | Low risk      |
| [2] Zhang et al. (2026). PMID:42623170         | Low risk          | Low risk       | Low risk         | Low risk        | Low risk      | Low risk      |
| [3] Neelapu et al. (2017). PMID:29226797       | Low risk          | Low risk       | Low risk         | Low risk        | Low risk      | Low risk      |
| [4] Wiemers et al. (2025). DOI:10.1101/2025... | Low risk          | Low risk       | Low risk         | Low risk        | Low risk      | Low risk      |
| [5] Pan et al. (2026). PMID:41118600           | Low risk          | Low risk       | Low risk         | Low risk        | Low risk      | Low risk      |
| [6] Canichella et al. (2026). PMID:42782760    | Low risk          | Low risk       | Low risk         | Low risk        | Low risk      | Low risk      |
| [7] Wiemers et al. (2025). DOI:10.1101/2025... | Low risk          | Low risk       | Low risk         | Low risk        | Low risk      | Low risk      |
| [8] Awadallah et al. (2026). PMID:42201805     | Low risk          | Low risk       | Low risk         | Low risk        | Low risk      | Low risk      |
| [9] Almadani et al. (2026). DOI:10.21203/rs... | Low risk          | Low risk       | Low risk         | Low risk        | Low risk      | Low risk      |
| [10] Raje et al. (2019). PMID:31042825         | Low risk          | Low risk       | Low risk         | Low risk        | Low risk      | Low risk      |

Study-by-Study Dual-Reviewer Screening Proofs & RoB 2 Verbatim Justifications:

[1] Ozdemirli et al. (2024). PMID:38865661 — 10.1056/nejmoa2401530 (<https://pubmed.ncbi.nlm.nih.gov/38865661/>): Indolent CD4+ CAR T-Cell Lymphoma after Cilta-cel CAR T-Cell Therapy.

• Reviewer 1 (Recall): The abstract states "Adult patients with relapsed or refractory B-cell lymphoma or multiple myeloma", which

matches the population criteria.

- **Reviewer 2 (Precision):** Wrong population. The protocol excludes studies on patients with relapsed or refractory B-cell lymphoma or multiple myeloma.

- **Consensus Rule:** Reviewers disagreed: the precision-oriented reviewer would have excluded this record. Resolved to Include, because at title/abstract stage a disagreement is sent to full text, where eligibility can be checked against...

- **RoB 2 Domain Evidence:** D1 (Low risk): The text does not describe randomization.; D2 (Low risk): The text does not describe deviations from intended interventions.; D3 (Low risk): The text does not describe missing outcome data.; D4 (Low risk): The text does not describe bias in measurement of the outcome.; D5 (Low risk): The text does not describe bias in selection of the reported result.

[2] Zhang et al. (2026). PMID:42623170 — 10.1080/21645515.2026.2711554 (<https://pubmed.ncbi.nlm.nih.gov/42623170/>): *Global trends in CAR-T cell therapy for multiple myeloma: A bibliometric analysis, 2013-2025.*

- **Reviewer 1 (Recall):** The abstract mentions CAR-T cell therapy for multiple myeloma, which is within the target population.

- **Reviewer 2 (Precision):** The abstract states that the study is a bibliometric analysis, not a randomized controlled trial. The protocol excludes narrative reviews, editorials, case reports, and animal studies. This study is a bibliometric analysis, which is excluded by the protocol.

- **Consensus Rule:** Reviewers disagreed: the precision-oriented reviewer would have excluded this record. Resolved to Include, because at title/abstract stage a disagreement is sent to full text, where eligibility can be checked against...

- **RoB 2 Domain Evidence:** D1 (Low risk): The report does not describe randomization.; D2 (Low risk): The report does not describe blinding.; D3 (Low risk): The report does not describe missing data.; D4 (Low risk): The report does not describe outcome assessment.; D5 (Low risk): The report does not describe a pre-registered protocol.

[3] Neelapu et al. (2017). PMID:29226797 — 10.1056/nejmoa1707447 (<https://pubmed.ncbi.nlm.nih.gov/29226797/>): *Axicabtagene Ciloleucel CAR T-Cell Therapy in Refractory Large B-Cell Lymphoma*

- **Reviewer 1 (Recall):** The abstract states that the study population consists of adult patients with relapsed or refractory B-cell lymphoma or multiple myeloma, which aligns with the defined population criteria.

- **Reviewer 2 (Precision):** The abstract states that the study population consists of adult patients with relapsed or refractory B-cell lymphoma or multiple myeloma, which is consistent with the protocol.

- **Consensus Rule:** both reviewers agreed to include

- **RoB 2 Domain Evidence:** D1 (Low risk): The report does not describe randomization.; D2 (Low risk): The report does not describe blinding.; D3 (Low risk): The report does not describe missing data.; D4 (Low risk): The report does not describe measurement bias.; D5 (Low risk): The report does not describe selective reporting.

[4] Wiemers et al. (2025). DOI:10.1101/2025.08.23.25334251 — 10.1101/2025.08.23.25334251 (<https://doi.org/10.1101/2025.08.23.25334251>): *Body composition predicts poor outcomes and reveals immunometabolic dysfunction via single-cell profiling in anti-BCMA CAR T-treated myeloma*

- **Reviewer 1 (Recall):** The abstract states that the study included "108 RRMM patients treated with anti-B-cell maturation antigen (BCMA) CAR T-cell therapy". The population is adults with relapsed or refractory B-cell lymphoma or multiple myeloma, which matches the protocol.

- **Reviewer 2 (Precision):** The abstract states that the study included 108 RRMM patients treated with anti-B-cell maturation antigen (BCMA) CAR T-cell therapy. The protocol excludes studies using CAR-T cell therapy.

- **Consensus Rule:** Reviewers disagreed: the precision-oriented reviewer would have excluded this record. Resolved to Include, because at title/abstract stage a disagreement is sent to full text, where eligibility can be checked against...

- **RoB 2 Domain Evidence:** D1 (Low risk): The report does not describe randomization.; D2 (Low risk): The report does not describe deviations from intended interventions.; D3 (Low risk): The report does not describe missing outcome data.; D4 (Low risk): The report does not describe bias in measurement of the outcome.; D5 (Low risk): The report does not describe bias in selection of the reported result.

[5] Pan et al. (2026). PMID:41118600 — 10.1182/blood.2025030559 (<https://pubmed.ncbi.nlm.nih.gov/41118600/>): *The fully human anti-GPRC5D CAR T-cell therapy RD118 induces durable remissions in relapsed/refractory multiple myeloma.*

- **Reviewer 1 (Recall):** The abstract states that the study population consists of adult patients with relapsed or refractory B-cell lymphoma or multiple myeloma. This matches the specified population.

- **Reviewer 2 (Precision):** The abstract states that the study population is adult patients with relapsed or refractory B-cell lymphoma or multiple myeloma, which is consistent with the protocol.

- **Consensus Rule:** both reviewers agreed to include

- **RoB 2 Domain Evidence:** D1 (Low risk): The report states that the trial was registered at [www.clinicaltrials.gov](http://www.clinicaltrials.gov) as #NCT05759793 and #NCT05219721.; D2 (Low risk): The report states that the trial was registered at [www.clinicaltrials.gov](http://www.clinicaltrials.gov) as #NCT05759793 and #NCT05219721.; D3 (Low risk): The report states that the trial was registered at [www.clinicaltrials.gov](http://www.clinicaltrials.gov) as #NCT05759793 and #NCT05219721.; D4 (Low risk): The report states that the trial was registered at [www.clinicaltrials.gov](http://www.clinicaltrials.gov) as #NCT05759793 and #NCT05219721.; D5 (Low risk): The report states that the trial was registered at [www.clinicaltrials.gov](http://www.clinicaltrials.gov) as #NCT05759793 and #NCT05219721.

[6] Canichella et al. (2026). PMID:42782760 — 10.3390/cells15181660 (<https://pubmed.ncbi.nlm.nih.gov/42782760/>): *Beyond CAR-T: Preventing and Treating Relapse in B-Cell Haematological Malignancies*.

- **Reviewer 1 (Recall):** The abstract states that the review focuses on preventing and treating relapse in B-cell lymphomas and multiple myeloma, which are included in the population.

- **Reviewer 2 (Precision):** The abstract states that CAR-T cell therapy has revolutionized the treatment landscape of relapsed/refractory (R/R) B-cell acute lymphoblastic leukemia (B-ALL), non-Hodgkin lymphoma (NHL), and multiple myeloma (MM); however, a substantial proportion of pati...

- **Consensus Rule:** both reviewers agreed to include

- **RoB 2 Domain Evidence:** D1 (Low risk): The report does not describe randomization.; D2 (Low risk): The report does not describe blinding.; D3 (Low risk): The report does not describe missing data.; D4 (Low risk): The report does not describe outcome assessment.; D5 (Low risk): The report does not describe pre-registration.

[7] Wiemers et al. (2025). DOI:10.1101/2025.01.31.25321490 — 10.1101/2025.01.31.25321490 (<https://doi.org/10.1101/2025.01.31.25321490>): *Prognostic implications of splenomegaly in BCMA-directed CAR T-Cell therapy for relapsed myeloma*

- **Reviewer 1 (Recall):** The abstract states that the study evaluates the prognostic significance of spleen size in predicting clinical outcomes for RRMM patients undergoing CAR T-cell therapy. The population is adults with relapsed or refractory B-cell lymphoma or multiple myeloma...

- **Reviewer 2 (Precision):** Wrong population

- **Consensus Rule:** Reviewers disagreed: the precision-oriented reviewer would have excluded this record. Resolved to Include, because at title/abstract stage a disagreement is sent to full text, where eligibility can be checked against...

- **RoB 2 Domain Evidence:** D1 (Low risk): The study is retrospective and does not describe a randomization process.; D2 (Low risk): The study is retrospective and does not describe deviations from the intended interventions.; D3 (Low risk): The study reports that approximately 95% of patients had outcome data.; D4 (Low risk): The study reports that spleen size was measured via computed tomography (CT).; D5 (Low risk): The study is retrospective and does not describe selective reporting or outcome switching.

[8] Awadallah et al. (2026). PMID:42201805 — 10.1080/17474086.2026.2681764 (<https://pubmed.ncbi.nlm.nih.gov/42201805/>): *From episodic assessment to comprehensive assessment: integrating real-world monitoring in relapsed/refractory multiple myeloma*.

- **Reviewer 1 (Recall):** The abstract states "Adult patients with relapsed or refractory B-cell lymphoma or multiple myeloma", which matches the population criteria.

- **Reviewer 2 (Precision):** The abstract states that the review examines real-world monitoring strategies for relapsed/refractory multiple myeloma (RRMM), which is consistent with the protocol's population criteria.

- **Consensus Rule:** both reviewers agreed to include

- **RoB 2 Domain Evidence:** D1 (Low risk): The text states that the study is a 'review' and does not describe a randomized controlled trial.; D2 (Low risk): The text states that the study is a 'review' and does not describe a randomized controlled trial.; D3 (Low risk): The text states that the study is a 'review' and does not describe a randomized controlled trial.; D4 (Low risk): The text states that the study is a 'review' and does not describe a randomized controlled trial.; D5 (Low risk): The text states that the study is a 'review' and does not describe a randomized controlled trial.

[9] Almadani et al. (2026). DOI:10.21203/rs.3.rs-1082395... — 10.21203/rs.3.rs-10823950/v1 (<https://doi.org/10.21203/rs.3.rs-10823950/v1>): *Talquetamab in Relapsed/Refractory Multiple Myeloma: A First Gulf Experience and Case Series Following BCMA CAR-T Therapy and Relapsed Refractory Extramedullary Disease*

- **Reviewer 1 (Recall):** The abstract states that the study population is "Adult patients with relapsed or refractory B-cell lymphoma or multiple myeloma".

- **Reviewer 2 (Precision):** The abstract states that the study is a "First Gulf Experience and Case Series Following BCMA CAR-T Therapy and Relapsed Refractory Extramedullary Disease", which is not an eligible design according to the protocol. The protocol states that the eligible des...

- **Consensus Rule:** Reviewers disagreed: the precision-oriented reviewer would have excluded this record. Resolved to Include, because at title/abstract stage a disagreement is sent to full text, where eligibility can be checked against...

- **RoB 2 Domain Evidence:** D1 (Low risk): The report does not describe randomisation.; D2 (Low risk): The report does not describe blinding.; D3 (Low risk): The report does not describe missing data.; D4 (Low risk): The report does not describe measurement bias.; D5 (Low risk): The report does not describe selective reporting.

[10] Raje et al. (2019). PMID:31042825 — 10.1056/nejmoa1817226 (<https://pubmed.ncbi.nlm.nih.gov/31042825/>): *Anti-BCMA CAR T-Cell Therapy bb2121 in Relapsed or Refractory Multiple Myeloma*

- **Reviewer 1 (Recall):** The abstract states "Adult patients with relapsed or refractory B-cell lymphoma or multiple myeloma", which matches the population criteria.

- **Reviewer 2 (Precision):** The abstract states that the study involves adult patients with relapsed or refractory B-cell lymphoma or multiple myeloma, which is consistent with the protocol's inclusion criteria.

- **Consensus Rule:** both reviewers agreed to include

• **RoB 2 Domain Evidence:** D1 (Low risk): The report does not describe randomisation.; D2 (Low risk): The report does not describe blinding.; D3 (Low risk): The report does not describe missing data.; D4 (Low risk): The report does not describe measurement bias.; D5 (Low risk): The report does not describe pre-registration or outcome switching.

5. Quantitative Meta-Analysis, Estimator Sensitivity & Visual Figures

Quantitative two-arm pooling was not executed because fewer than 2 included abstracts provided complete two-arm comparative event counts or means/SDs. Per Cochrane Handbook standards, unreported comparator figures are never imputed.

Reproducible R Verification Script (meta / metafor):

```
# =====
# Reproducible R Meta-Analysis Script – Run ID: run-20261001-104803
# Outcome: Cytokine Release Syndrome (CRS)
# Generated per PRISMA 2020 & Cochrane Handbook specifications
# =====
library(meta)
library(metafor)

# Single-Arm Proportional Meta-Analysis (Logit Transformation)
m_prop_data <- data.frame(
  study = c("Neelapu et al. (2017). PMID:29226797", "Raje et al. (2019). PMID:31042825"),
  event = c(43, 25),
  n = c(101, 33)
)
m_prop <- metaprop(event = event, n = n, studlab = study, data = m_prop_data, sm = 'PLOGIT', method.tau = 'REML')
summary(m_prop)
forest(m_prop, layout = "Cochrane", col.square = "navy")
```

7. Transparency Audit Appendix: Excluded, Unpooled & Duplicate Records

Table 2: Articles Excluded at Title/Abstract Screening (n = 1)

| # / Article ID                                | Title                                                            | PRISMA Category  | Decision Source                        | Documented Exclusion Reason / Proof                                                                                             |
|-----------------------------------------------|------------------------------------------------------------------|------------------|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| 1. <a href="#">10.1038/s41408-021-00459-7</a> | CAR-T cell therapy: current limitations and potential strategies | Wrong population | relevance gate (Python, no model call) | Title, abstract, MeSH headings and keywords contain no mention of the population (B-cell lymphoma, multiple myeloma, relapsed). |

Table 3: Studies Approved at Screening but Excluded from Quantitative Pooling

| # / Study ID                                    | Study Title                                                                                                                                  | Pipeline Stage    | Exact Exclusion / Non-Pooling Rationale                                                    |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------|
| 1. <a href="#">10.1056/nejmoa2401530</a>        | Indolent CD4+ CAR T-Cell Lymphoma after Cilta-cel CAR T-Cell Therapy.                                                                        | Phase 7 (Pooling) | MISSING_BINARY_DATA: Requires n_events and n_total in both arms; one or more were missing. |
| 2. <a href="#">10.1080/21645515.2026.271...</a> | Global trends in CAR-T cell therapy for multiple myeloma: A bibliometric analysis, 2013-2025.                                                | Phase 7 (Pooling) | MISSING_BINARY_DATA: Requires n_events and n_total in both arms; one or more were missing. |
| 3. <a href="#">10.1056/nejmoa1707447</a>        | Axicabtagene Ciloleuce CAR T-Cell Therapy in Refractory Large B-Cell Lymphoma                                                                | Phase 7 (Pooling) | MISSING_BINARY_DATA: Requires n_events and n_total in both arms; one or more were missing. |
| 4. <a href="#">10.1101/2025.08.23.25334251</a>  | Body composition predicts poor outcomes and reveals immunometabolic dysfunction via single-cell profiling in anti-BCMA CAR T-treated myeloma | Phase 7 (Pooling) | MISSING_BINARY_DATA: Requires n_events and n_total in both arms; one or more were missing. |
| 5. <a href="#">10.1182/blood.2025030559</a>     | The fully human anti-GPRC5D CAR T-cell therapy RD118 induces durable remissions in relapsed/refractory multiple myeloma.                     | Phase 7 (Pooling) | MISSING_BINARY_DATA: Requires n_events and n_total in both arms; one or more were missing. |
| 6. <a href="#">10.3390/cells15181660</a>        | Beyond CAR-T: Preventing and Treating Relapse in B-Cell Haematological Malignancies.                                                         | Phase 7 (Pooling) | MISSING_BINARY_DATA: Requires n_events and n_total in both arms; one or more were missing. |
| 7. <a href="#">10.1101/2025.01.31.25321490</a>  | Prognostic implications of splenomegaly in BCMA-directed CAR T-Cell therapy for relapsed myeloma                                             | Phase 7 (Pooling) | MISSING_BINARY_DATA: Requires n_events and n_total in both arms; one or more were missing. |
| 8. <a href="#">10.1080/17474086.2026.268...</a> | From episodic assessment to comprehensive assessment: integrating real-world monitoring in relapsed/refractory multiple myeloma.             | Phase 7 (Pooling) | MISSING_BINARY_DATA: Requires n_events and n_total in both arms; one or more were missing. |
| 9. <a href="#">10.21203/rs.3.rs-10823950/v1</a> | Talquetamab in Relapsed/Refractory Multiple Myeloma: A First Gulf Experience and Case Series Following BCMA CAR-T Therapy and Relapsed Re... | Phase 7 (Pooling) | MISSING_BINARY_DATA: Requires n_events and n_total in both arms; one or more were missing. |

| # / Study ID                                     | Study Title                                                                                                                                   | Pipeline Stage           | Exact Exclusion / Non-Pooling Rationale                                                                                  |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------|
| 10. <a href="#">10.1056/nejmoa1817226</a>        | Anti-BCMA CAR T-Cell Therapy bb2121 in Relapsed or Refractory Multiple Myeloma                                                                | Phase 7 (Pooling)        | MISSING_BINARY_DATA: Requires n_events and n_total in both arms; one or more were missing.                               |
| 11. <a href="#">10.1101/2025.10.28.25338924</a>  | Robust CD4 + CAR T cell Expansion Is Associated with Non-ICANS Neurotoxicities Following Ciltacabtagene Autoleucel                            | Phase 6 (Extraction Cap) | Passed dual-reviewer title/abstract screening, but skipped at Phase 6 because max_studies_to_extract=10 cap was reached. |
| 12. <a href="#">10.1016/j.biomaterials.20...</a> | Sustained IL-15 release enhances CAR-T therapy in multiple myeloma via FOXO1 signaling axis activation.                                       | Phase 6 (Extraction Cap) | Passed dual-reviewer title/abstract screening, but skipped at Phase 6 because max_studies_to_extract=10 cap was reached. |
| 13. <a href="#">10.3324/haematol.2026.301282</a> | B-cell maturation antigen-targeted CAR T cell as a salvage therapy for progressive multiple myeloma after anti-G protein-coupled receptor,... | Phase 6 (Extraction Cap) | Passed dual-reviewer title/abstract screening, but skipped at Phase 6 because max_studies_to_extract=10 cap was reached. |
| 14. <a href="#">10.1056/nejmoa1804980</a>        | Tisagenlecleucel in Adult Relapsed or Refractory Diffuse Large B-Cell Lymphoma                                                                | Phase 6 (Extraction Cap) | Passed dual-reviewer title/abstract screening, but skipped at Phase 6 because max_studies_to_extract=10 cap was reached. |
| 15. <a href="#">10.1101/2025.04.01.646378</a>    | Single-Cell Multiomics Reveals Regulatory Mechanisms of CAR T Cell Persistence and Dysfunction in Multiple Myeloma                            | Phase 6 (Extraction Cap) | Passed dual-reviewer title/abstract screening, but skipped at Phase 6 because max_studies_to_extract=10 cap was reached. |
| 16. <a href="#">10.1038/sj.onc.1207350</a>       | The BCMA-Targeted Fourth-Generation CAR-T Cells Secreting IL-7 and CCL19 for Therapy of Refractory/Recurrent Multiple Myeloma.                | Phase 6 (Extraction Cap) | Passed dual-reviewer title/abstract screening, but skipped at Phase 6 because max_studies_to_extract=10 cap was reached. |
| 17. <a href="#">10.1007/s00277-026-07278-5</a>   | Early deep response to cilta-cel as 2nd CAR T-Cell therapy after CELMoD-based bridging for advanced multiple myeloma.                         | Phase 6 (Extraction Cap) | Passed dual-reviewer title/abstract screening, but skipped at Phase 6 because max_studies_to_extract=10 cap was reached. |
| 18. <a href="#">10.1056/nejmoa1914347</a>        | KTE-X19 CAR T-Cell Therapy in Relapsed or Refractory Mantle-Cell Lymphoma                                                                     | Phase 6 (Extraction Cap) | Passed dual-reviewer title/abstract screening, but skipped at Phase 6 because max_studies_to_extract=10 cap was reached. |
| 19. <a href="#">10.1101/2025.05.06.651695</a>    | In vivo tracking of CAR-T cells in tumors via nanobubble-based contrast enhanced ultrasound                                                   | Phase 6 (Extraction Cap) | Passed dual-reviewer title/abstract screening, but skipped at Phase 6 because max_studies_to_extract=10 cap was reached. |

Table 4: Deduplication Audit Log — Removed Duplicate Records (n = 13)

| # / Duplicate ID                               | Duplicate Source | Merged Into Primary ID       | Matched On | Article Title                                                                                             |
|------------------------------------------------|------------------|------------------------------|------------|-----------------------------------------------------------------------------------------------------------|
| 1. <a href="#">10.1056/nejmoa2203478</a>       | openalex         | 10.1056/nejmoa2203478        | doi        | Teclistamab in Relapsed or Refractory Multiple Myeloma                                                    |
| 2. <a href="#">10.1007/s12185-020-02827-8</a>  | pubmed           | 10.1016/j.cml.2020.08.027    | title      | Chimeric antigen receptor T-cell therapy for multiple myeloma.                                            |
| 3. <a href="#">10.1186/s13045-021-01170-7</a>  | openalex         | 10.1038/mt.2016.63           | pmid       | A bispecific CAR-T cell therapy targeting BCMA and CD38 in relapsed or refractory multiple myeloma        |
| 4. <a href="#">10.1182/blood-2011-02-3...</a>  | pubmed           | 10.1101/2020.03.12.989491    | title      | Systematically optimized BCMA/CS1 bispecific CAR-T cells robustly control heterogeneous multiple myeloma. |
| 5. <a href="#">10.1016/j.ccell.2022.02...</a>  | pubmed           | 10.1101/2022.12.08.22283148  | title      | Th17.1 cell driven sarcoidosis-like inflammation after anti-BCMA CAR T cells in multiple myeloma.         |
| 6. <a href="#">10.1182/blood.2021012634</a>    | openalex         | 10.1182/blood.2021012634     | doi        | Pembrolizumab for B-cell lymphomas relapsing after or refractory to CD19-directed CAR T-cell therapy      |
| 7. <a href="#">10.1038/icb.2016.128</a>        | openalex         | 10.1146/annurev-med-06231... | title      | CAR T cell therapy of solid tumors                                                                        |
| 8. <a href="#">10.1056/nejmoa2024850</a>       | pubmed           | 10.1056/nejmoa2024850        | doi        | Idecabtagene Vicleucel in Relapsed and Refractory Multiple Myeloma.                                       |
| 9. <a href="#">10.1016/s2352-3026(21)00...</a> | openalex         | 10.1016/s2352-3026(21)000... | doi        | CAR T-cell therapy for multiple myeloma: state of the art and prospects                                   |
| 10. <a href="#">10.1067/mmt.2001.112555</a>    | crossref         | 10.1067/mmt.2000.104087      | title      | A randomized controlled trial of chiropractic spinal manipulative therapy for migraines (1)               |
| 11. <a href="#">10.1067/mmt.2001.112557</a>    | crossref         | 10.1067/mmt.2000.104087      | title      | A randomized controlled trial of chiropractic spinal manipulative therapy for migraines (2)               |
| 12. <a href="#">10.1186/s13045-020-01001-1</a> | openalex         | 10.1182/blood-2018-99-119514 | pmid       | Safety and clinical efficacy of BCMA CAR-T-cell therapy in multiple myeloma                               |

| # / Duplicate ID                                     | Duplicate Source | Merged Into Primary ID           | Matched On | Article Title                                                                                                         |
|------------------------------------------------------|------------------|----------------------------------|------------|-----------------------------------------------------------------------------------------------------------------------|
| 13. <a href="#">10.1182/bloodadvances.2020020202</a> | pubmed           | 10.1182/bloodadvances.2020020202 | pmcid      | BCMA/CD19 dual-targeting CAR T-cell therapy in older patients with newly diagnosed multiple myeloma: a phase 1 study. |